1. Adam is working in an IT company. He has been given a task to reduce the load of a system by killing some of the processes running in the Linux operating system. Which commands will he use to complete the given task with help of the following operations

- Kill processes by name.
- Kill a process based on the process name.
- Kill a single process at a time with the given process ID.

→

ps -ef - To display all runing process in system.

Killall process :- Kill all runing with given name.

pkill process name - Kill processes by matching their name or pattern.

Kill PID :- safety terminalls one specific process.

Kill 9 :- forcefuly kill a process used when it is stuck.

sleep 300 & :- To create a dummy process for testing.

Killall sleep :- to terminal all process with the same name.

2. Write a program for process creation using c.

- orphan process

→ orphan process :- create when the parent process terminate refer the child process. the child process continue excution & it adopted by the system.

→ nano orphan.c → To create & write a c program for orphan process
→ gcc orphan.c → o orphan → To compile the program
→ ./orphan → To excute the compiled program.

- zombie process.

→ zombie process: create when the child process terminate the parent process does not call wait() the child remain in the zombie state the parent exits.
→ nano zombie.c → To create the zombie process program.
→ gcc zombie.c -o zombie → To compile the zombie program
→ ./zombie → To excute to zombie process program.

3) create the process using fork() system call.
- child process creation.
- Parent process creation.
- PPID & PID

→ fork() → this is used to create a new process. After fork() ~~with~~ parent & child process excute independently.

→ nano fork.c → To create a c program using fork().

→ gcc fork.c -o fork → To compill the fork program.

→ ./fork → To excute the program & observe parent & child processes.